

UNIVERSITY OF PÉCS
NATIONAL LABORATORY
OF VIROLOGY

Aedes koreicus in Europe: invasion history, dispersal, and perspectives from ongoing research



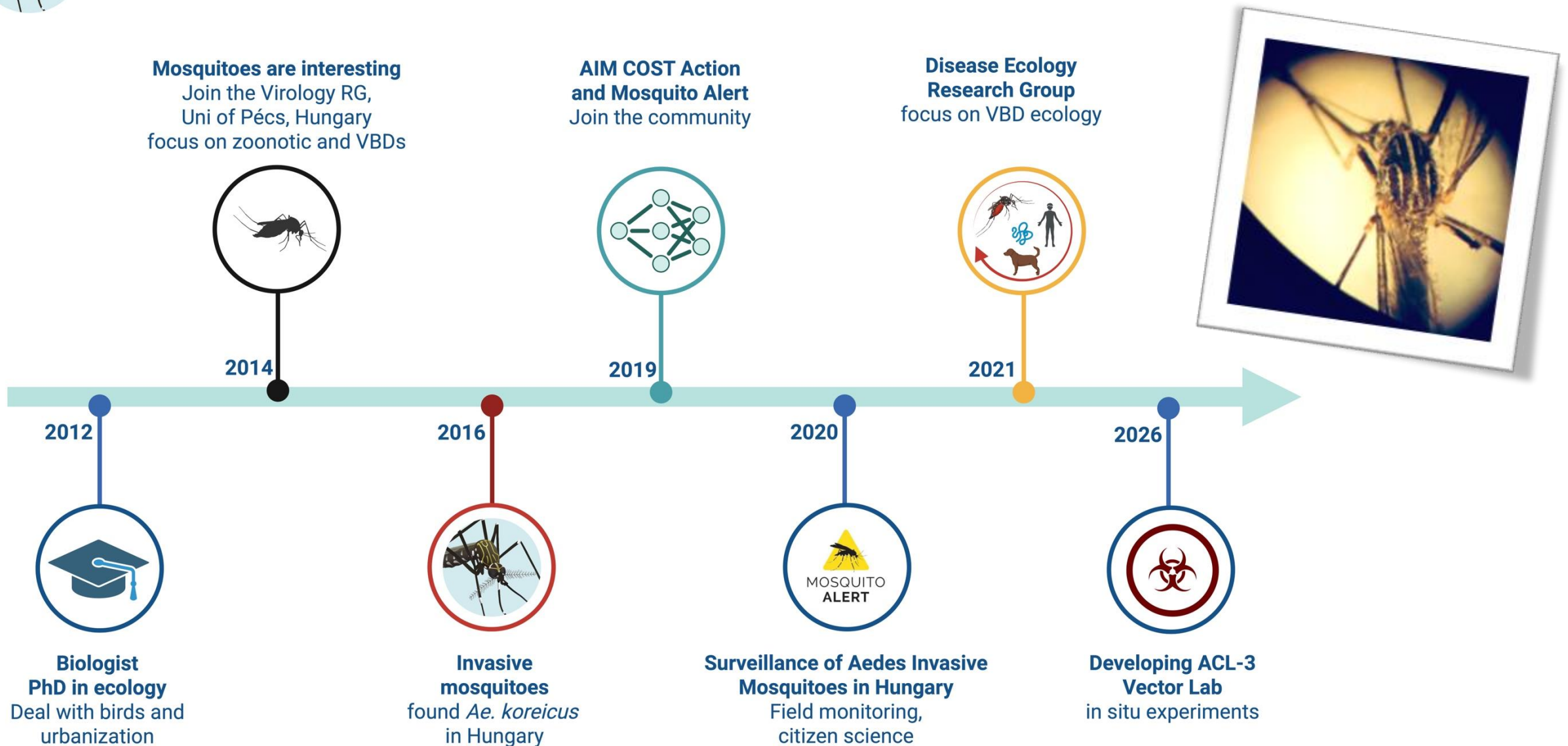
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My relationship with *Aedes koreicus* – love at first sight





Emergence and dispersal of *Aedes koreicus* in Europe

- Endemic in Asia, the first European population “probably” came from Korea, Jeju-Do
- Similar morphology to *Aedes japonicus* makes it harder to identify the species
- Lack of Europe-wide surveillance and/or more focus on *Aedes albopictus*
- Considering the introduction via global trade, international/intercontinental transport routes
- **Tracking the species’ spread across Europe is uncertain**

Reported outside its native region:

2008 – Belgium

2011 – Italy

2013 – Switzerland

Slovenia (retrospectively!)

2015 – Germany

2016 – Hungary

2018 – Austria

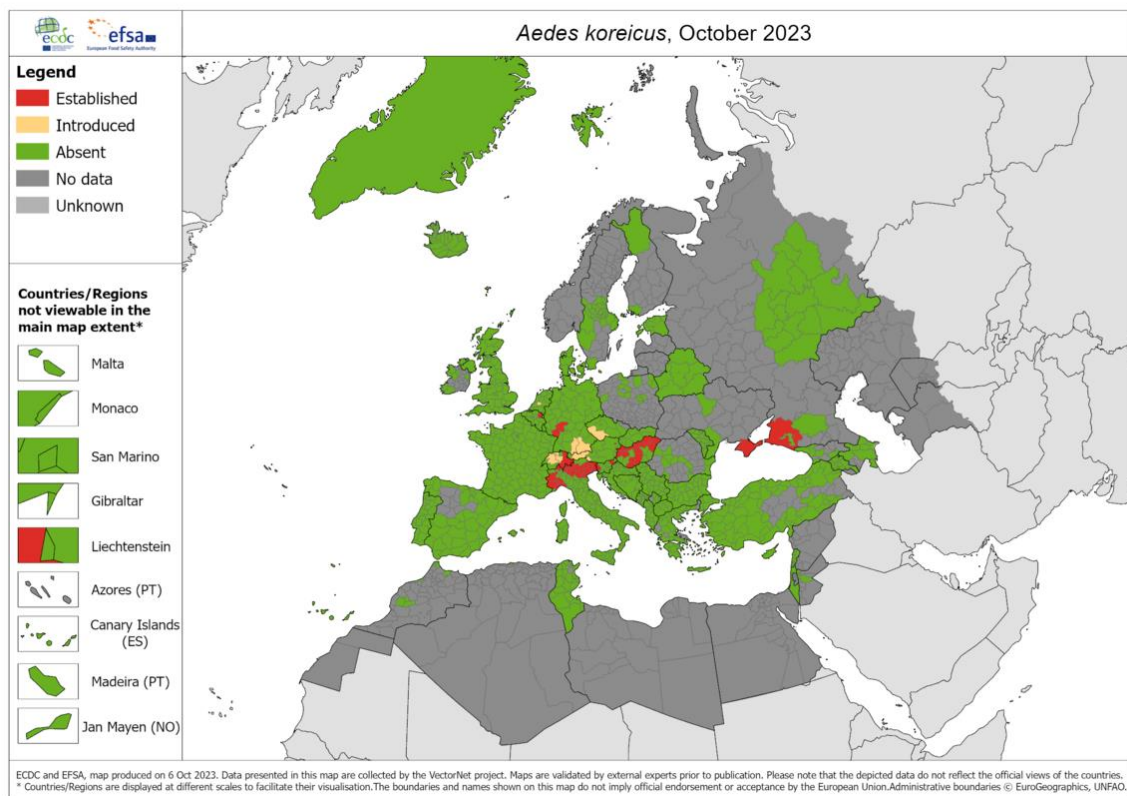
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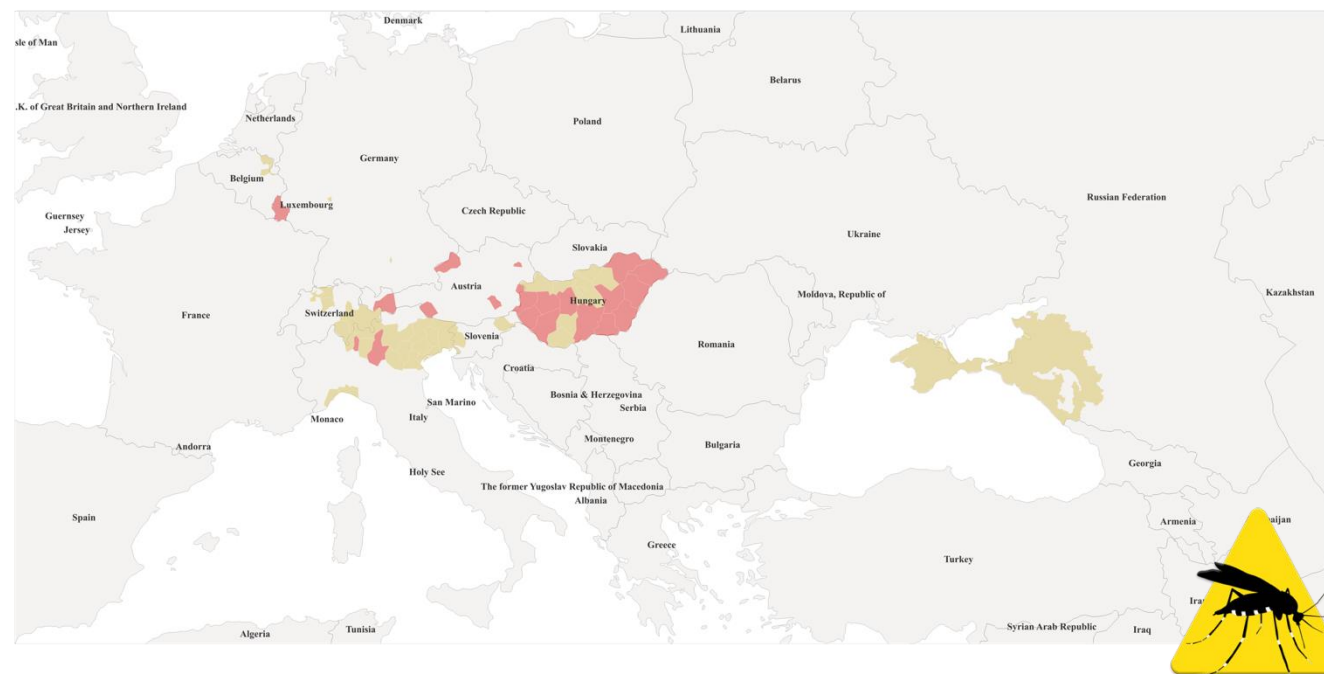
Current „known” distribution of *Aedes koreicus* in Europe

- Depends on surveillance efforts: field surveys and studies
- Since 2020, completed with citizen science data
- Limited by citizens' activity and efforts invested in advertising



The known distribution of *Aedes koreicus* in Europe

- Official data from the European Centre for Disease Prevention and Control at NUTS3 level (www.ecdc.europa.eu)
- Community-driven discoveries via Mosquito Alert citizen science program (www.mosquitoalert.com)

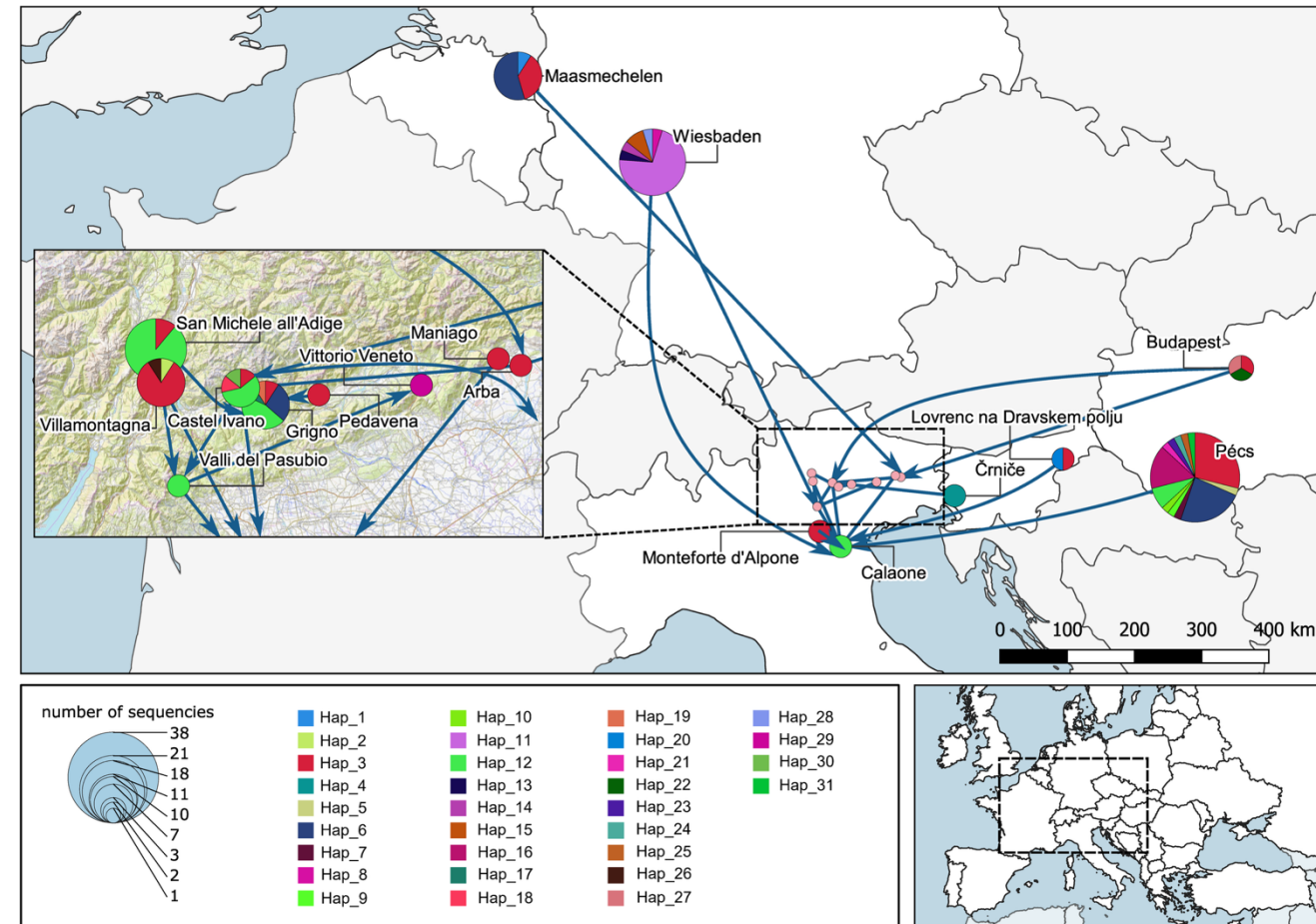
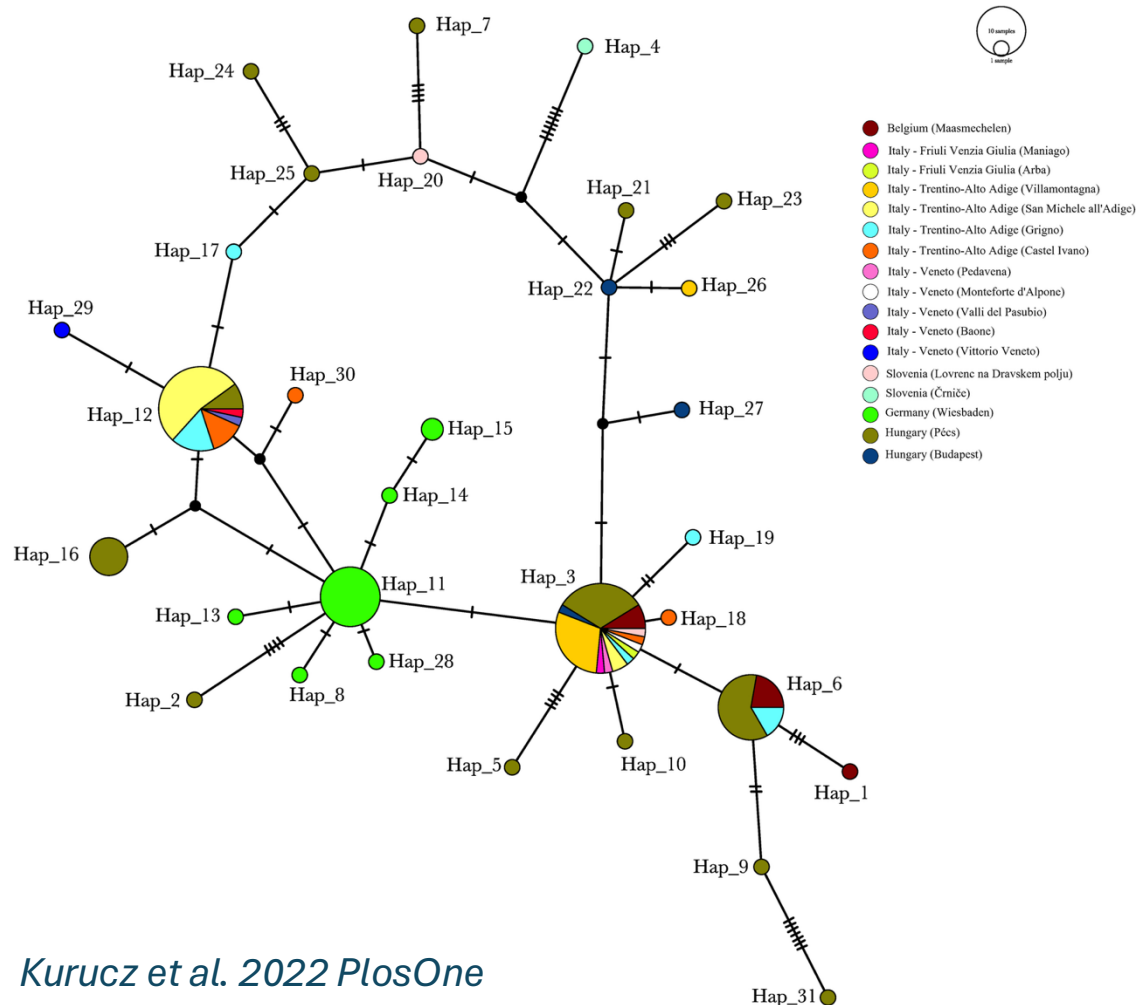


Sources: <https://ecdc.europa.eu/en/disease-vectors/surveillance-and-disease-data/mosquito-maps>, <https://map.mosquitoalert.com/en>



Pan-European genetic patterns of *Aedes koreicus*

- Haplotype network analysis (COI) – 4-5 major groups (31 haplotypes)
- Phylogeography – **multiple dispersal routes, multiple introductions to Europe**





Complete genome of *Aedes koreicus*

- Complete mitochondrial genome,
 - High-quality draft genome sequencing (1.1 Gbp long, 6099 contigs) from the Hungarian population
 - Annotation of genes probably involved in resistance to insecticides
- ↪
- Opens the possibility to understand genetic traits that facilitate the successful establishment and survival of the species
 - Form the **basis for further downstream genetic analyses** targeting mosquito control

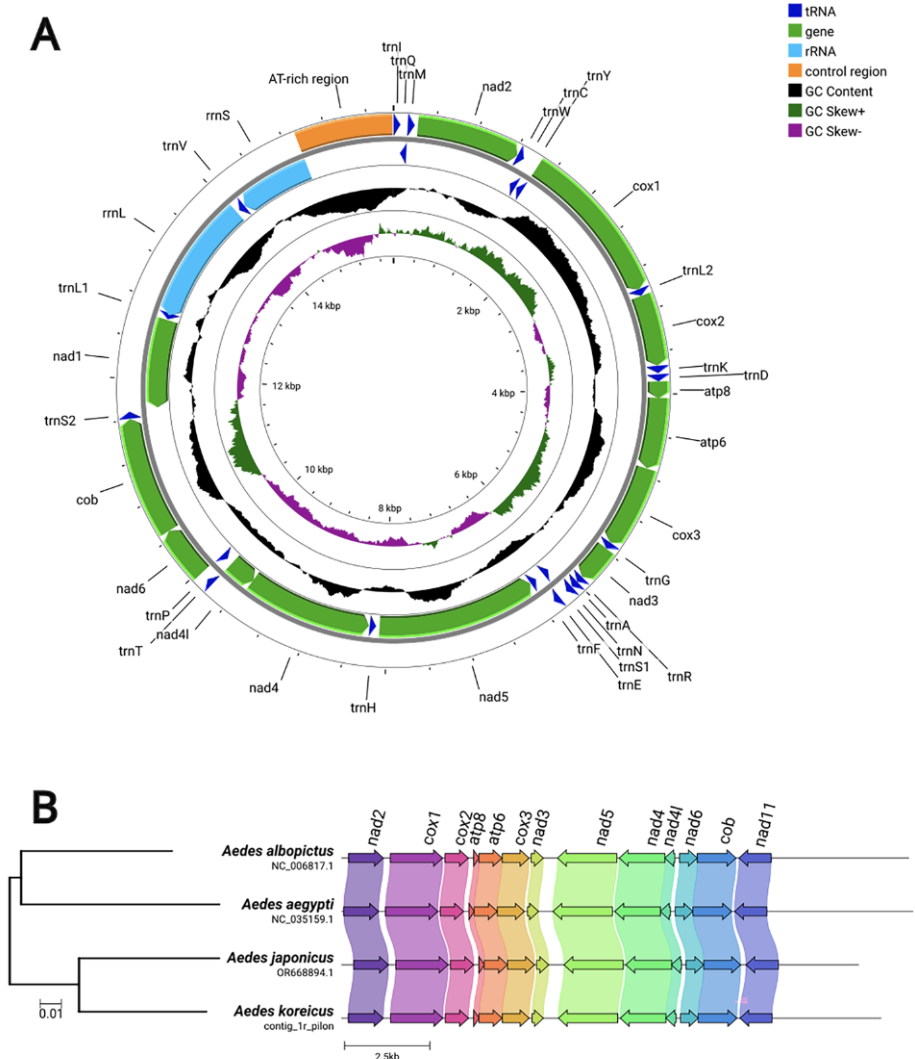


Figure 2. The circularized mitochondrial genome visualized by Proksee (A) and the structural comparison of the mitochondrial genomes of *Aedes* species with available complete mitochondrial genomes (B).



Morphological variability of *Aedes koreicus*

- Since 2019, morphological differences between collected specimens from the same population (Pécs, Hungary)

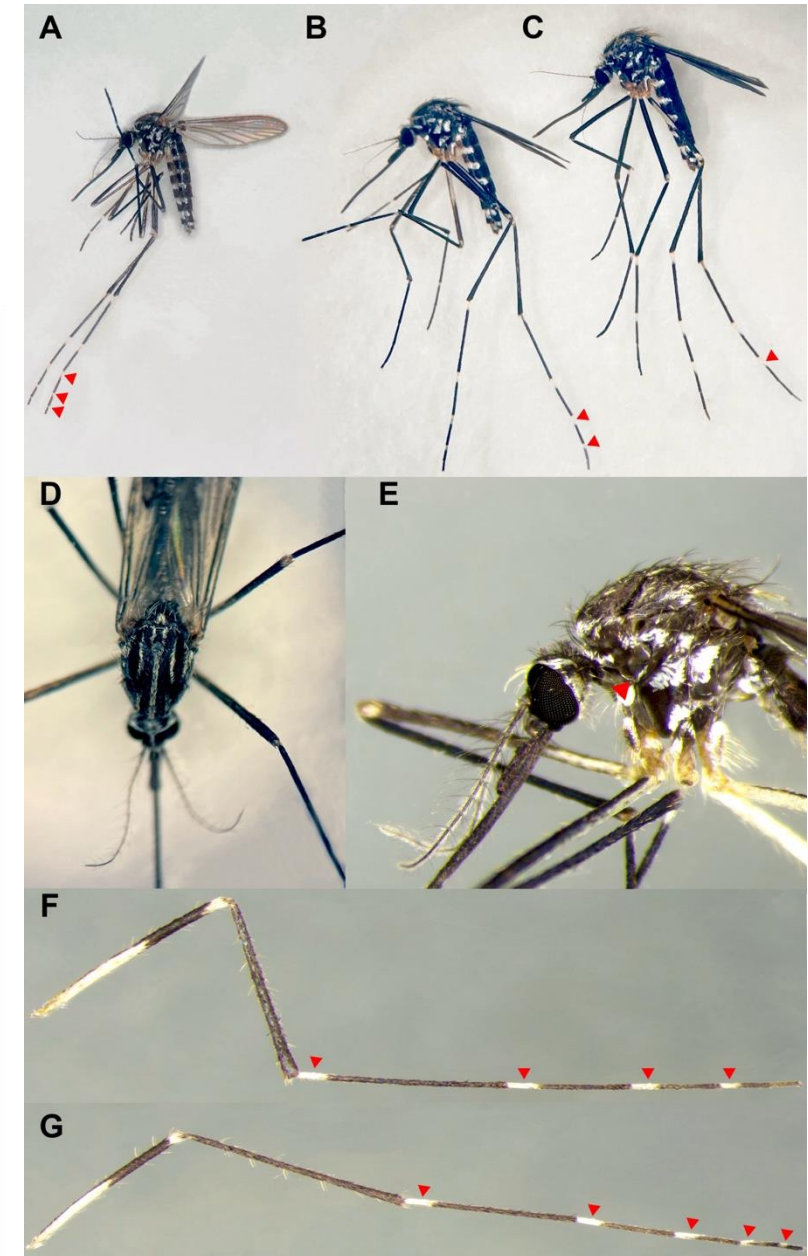
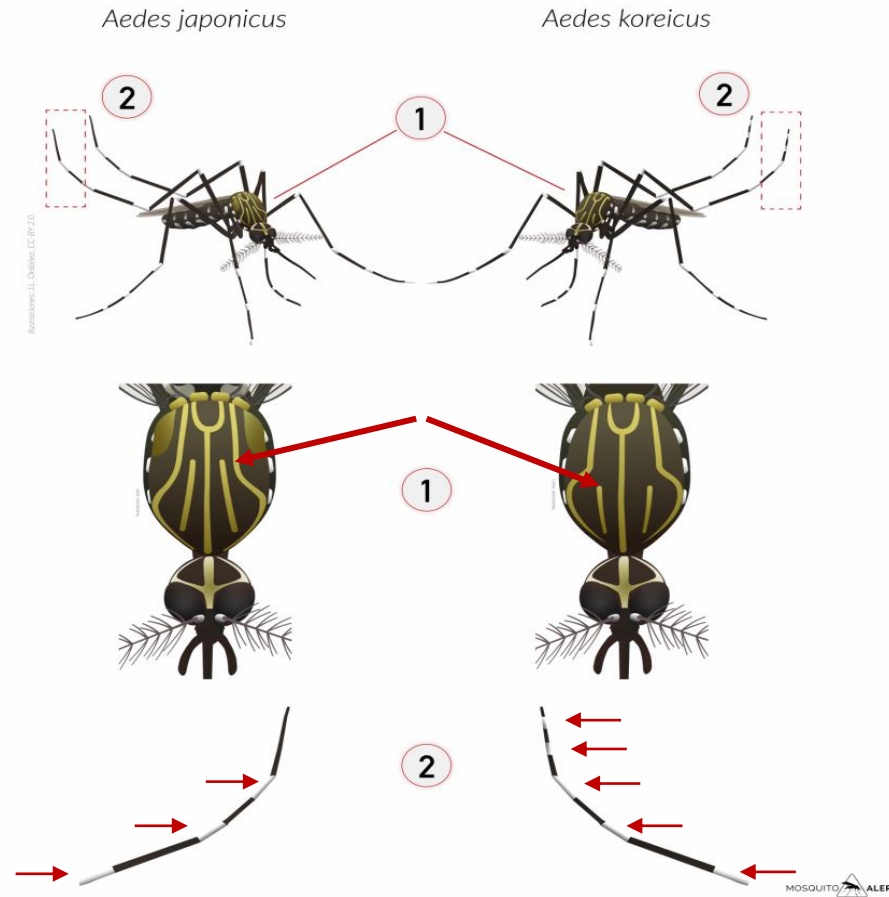
Guidelines for the surveillance of invasive mosquitoes in Europe

TECHNICAL REPORT

Figure 9: Main diagnostic morphological characters for adults of IMS

Figure 9: Main diagnostic morphological characters for adults of IMS. A. Thorax (scutum, dorsal side); B. Abdomen (dorsal side); C. Hind tarsus (last segments of the third leg). Males have a more hairy and slender abdomen. A. Thorax (scutum, dorsal side); B. Abdomen (dorsal side); C. Hind tarsus (last segments of the third leg). Males have a more hairy and slender abdomen.

<i>Ae. aegypti</i>	<i>Ae. albopictus</i>	<i>Ae. atropalpus</i>	<i>Ae. f. japonicus</i>	<i>Ae. koreicus</i>	<i>Ae. triseriatus</i>
A. Silver scales in a shape of a lyre	A. Median silver-scale line	A. Mixed yellow and dark scales, dark median stripe	A. 5 stripes of golden scales, submedians long	A. 5 stripes of golden scales, submedians short	A. Broad median dark-scale band, sides with silver scales
B. Pale lateral marks and thin basal bands	B. Thin basal pale bands, laterally enlarged	B. Pale basal bands	B. Pale lateral and median patches	B. Pale lateral and median patches	B. Pale lateral patches
C. Segment V entirely white	C. Segment V entirely white	C. Inter-articular pale bands S. V white	C. Segments IV & V black	C. Segments IV and V with basal pale bands	C. Tarsi entirely dark

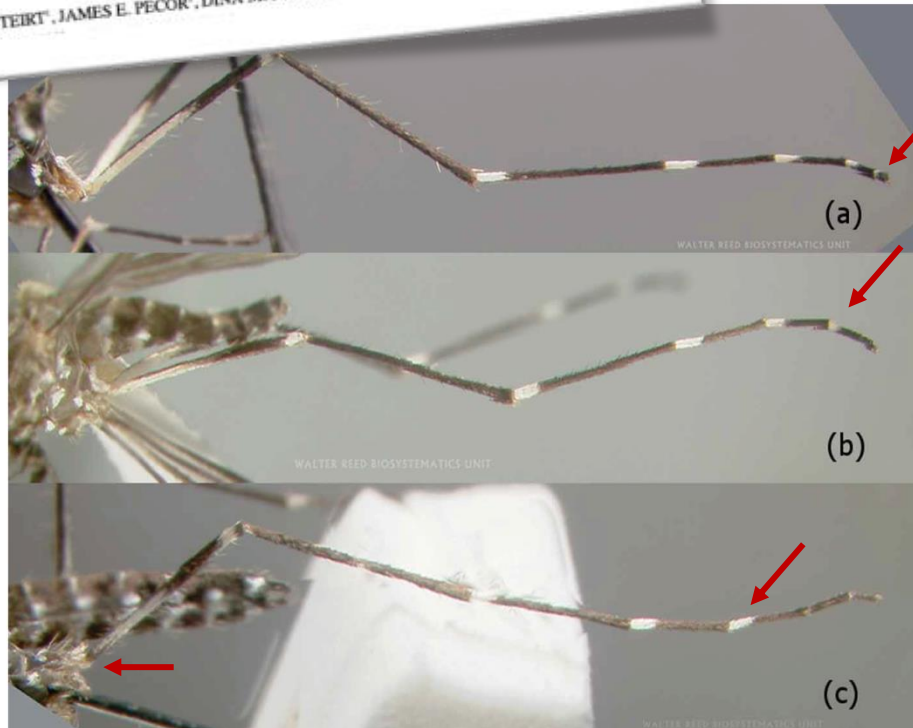




Morphological variability of *Aedes koreicus*



„Tanaka et al. (1979) mentioned a number of morphological differences between adult females collected on the Korean Peninsula and Jeju-do that correspond to the observed differences between Korean-type specimens and Belgian specimens.“



Ae. koreicus
„Jeju-Do“

Ae. koreicus
„Mainland“

Ae. japonicus

FIGURE 4. Differences in hindleg ornamentation of (a) *Aedes koreicus* from Belgium, (b) *Aedes koreicus* from peninsular Korea and (c) *Aedes j. japonicus* from Belgium.



Morphological variability of *Aedes koreicus*



„The newly discovered population in Germany shows different morphological characteristics from the populations in Belgium and Italy; it seems to originate from an independent introduction”

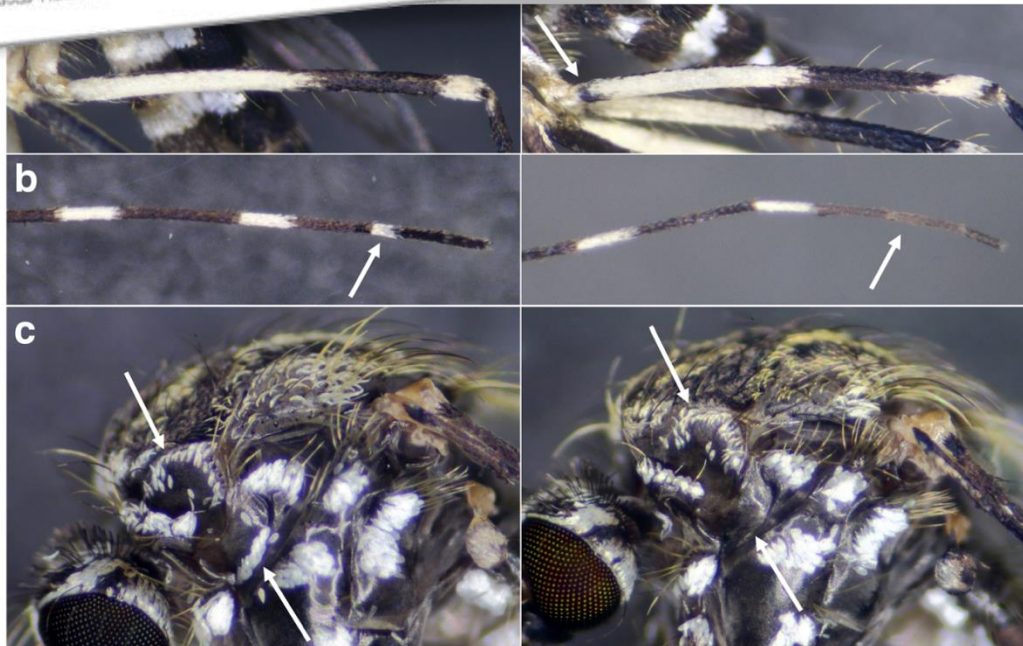


Fig. 2 Diagnostic characteristics of *Ae. koreicus* females (left) in comparison to *Ae. j. japonicus* (right). **a** Hind femur. The arrow shows the dark subbasal band in *Ae. j. japonicus*, which is missing in *Ae. koreicus*. **b** Hind tarsus with tarsomeres II-V. Arrows show hindtarsomere IV. **c** Lateral view of thorax. Arrows show the postpronotum and the subspiracular patch (missing in *Ae. j. japonicus*). The costa can also be seen

Reported outside its native region:

2008 – Belgium („*Jeju-Do*”)

2011 – Italy

2013 – Switzerland

Slovenia (retrospectively!)

2015 – Germany („*mainland Korea*”)

2016 – Hungary („*Jeju-Do*”)

2019 – Hungary („*mainland Korea*”)

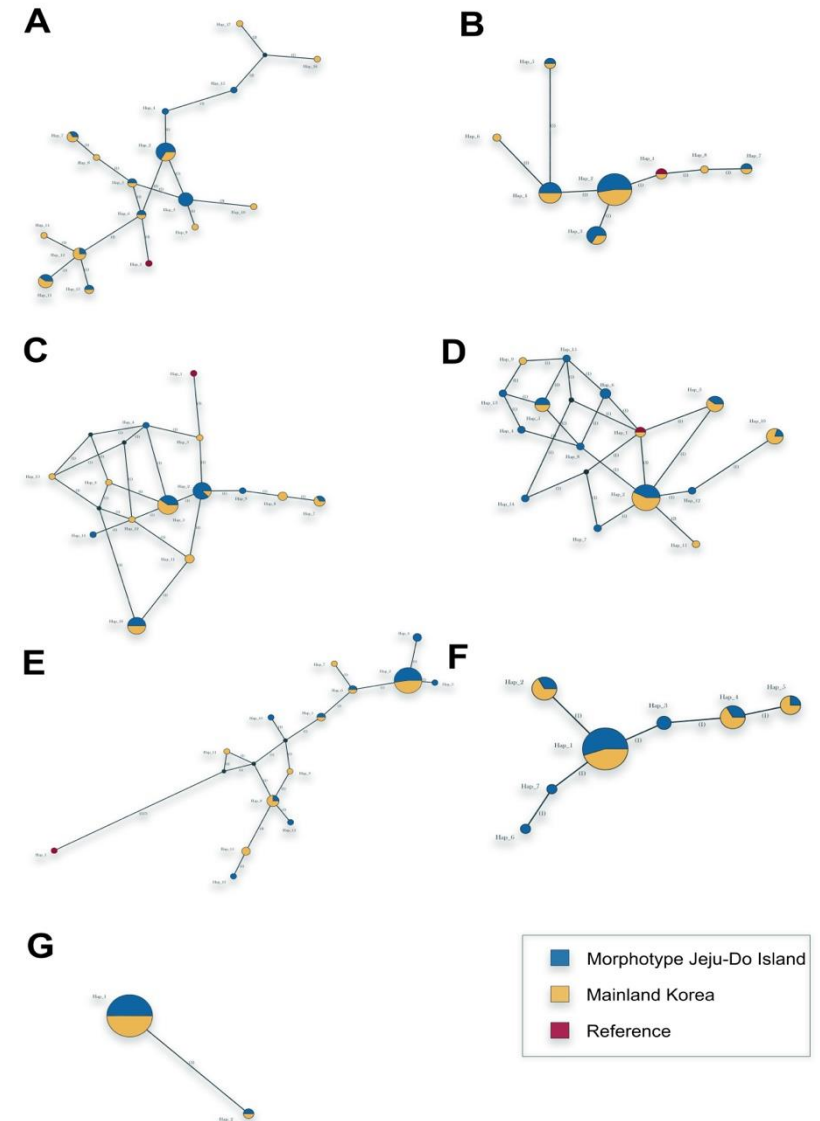
2018 – Austria

...



Morphological variability of *Aedes koreicus*

- 20 - 20 specimens of “Jeju-Do” and “mainland Korea” from the same population (location)
 - (reference species: *Ae. japonicus*)
 - Haplotype network analysis
 - Phylogenetic analyses (Maximum likelihood, Bayesian)
 - 7 different markers: COX1, COX2, COX3, ATP6, ND1, ND3, and ITS2
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- **No detectable mitochondrial or nuclear genetic differentiation** on the population level
 - Demonstrating the **co-occurrence of distinct morphological forms** within a single population

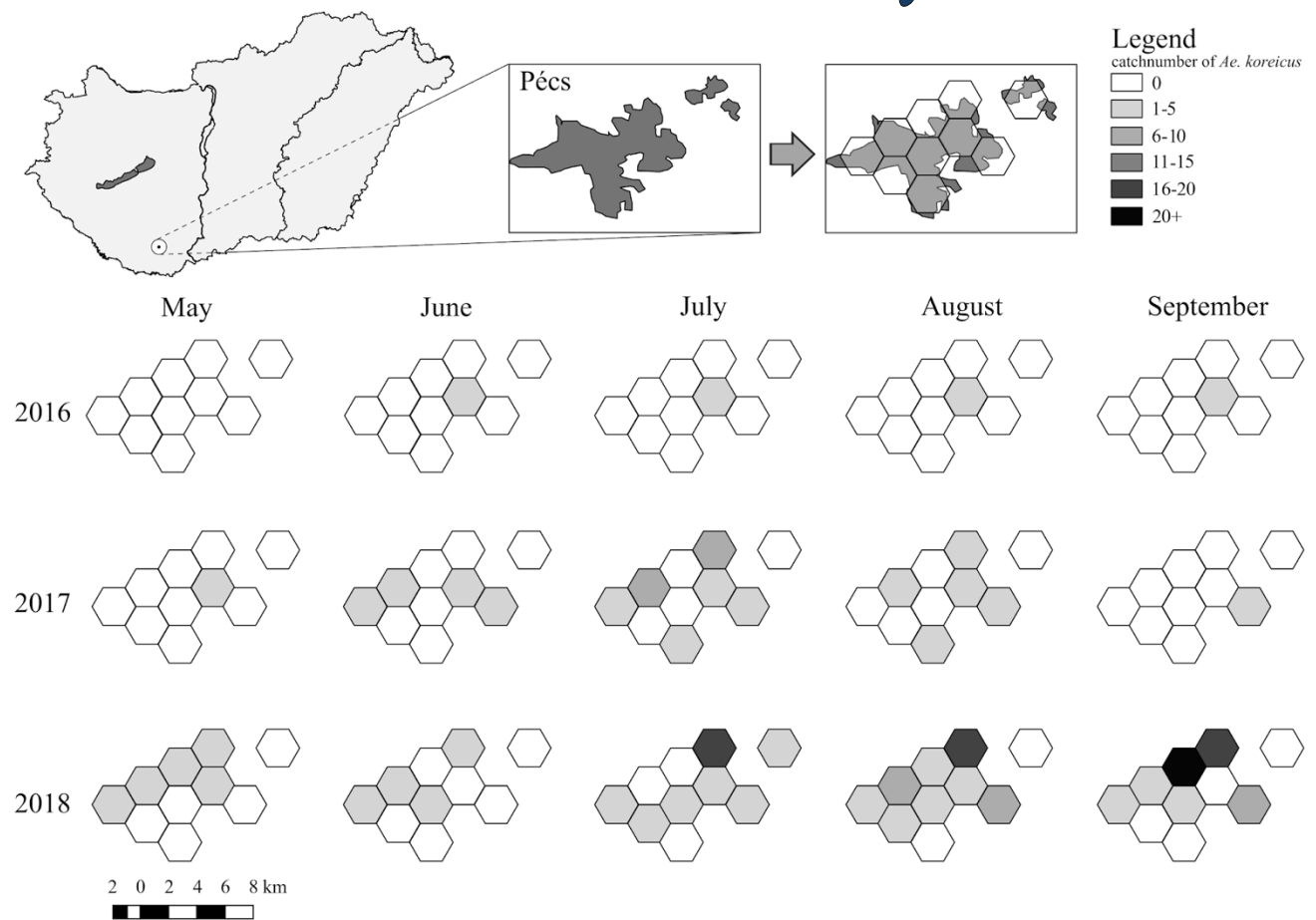
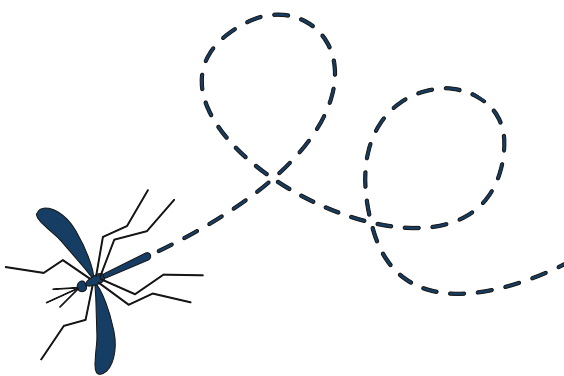




Invasion history of *Aedes koreicus* in Pécs, Hungary

Faunistical monitoring of mosquitoes since 2015

- First emergence of *Ae. koreicus* in Hungary (2016)

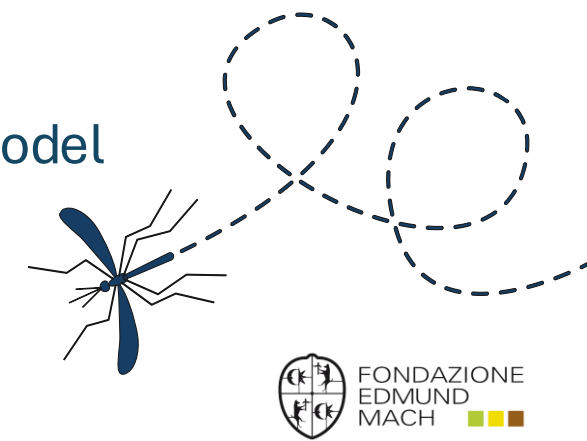




Invasion history of *Aedes koreicus* in Pécs, Hungary

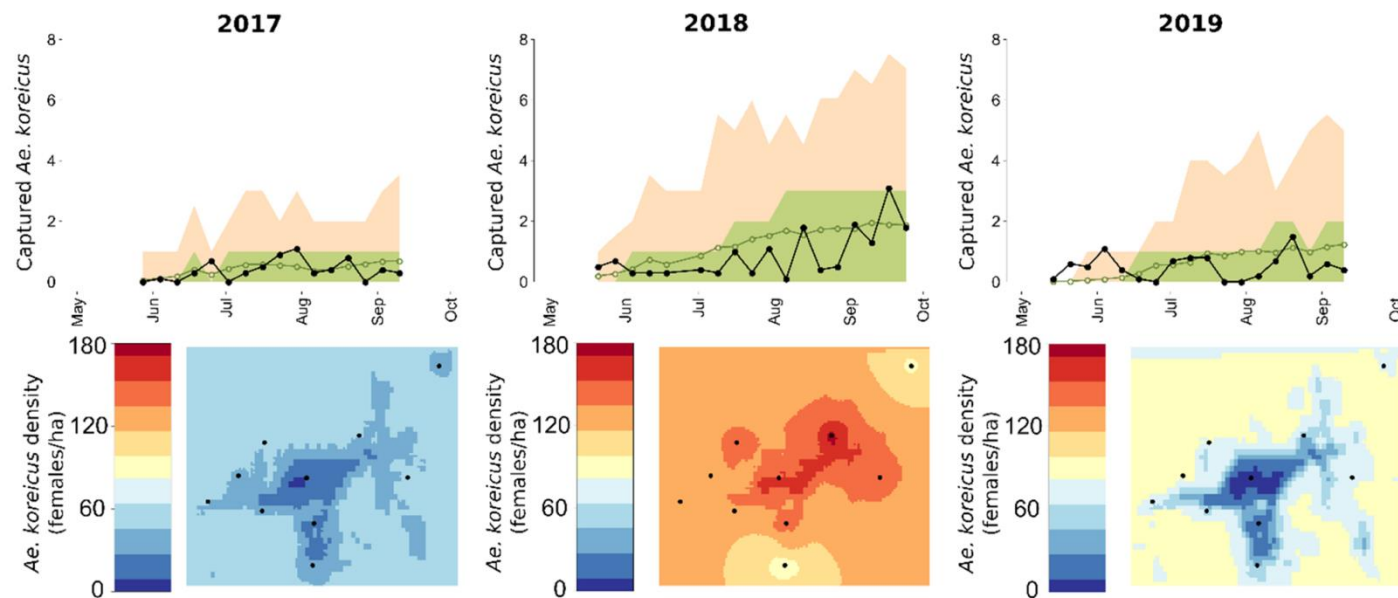
Applied a previously developed temperature-dependent mathematical model
(*Marini et al. 2019 Parasites and Vectors*)

- replicates the temporal dynamics observed with trapping
- negative relationship with summer temperature – implying **that too warm conditions might decrease *Ae. koreicus* survival**
- slightly negative correlation with human density – suggesting ***Ae. koreicus* can complete its life cycle feeding on animals other than humans**



- high variability in trap captures
- limited number of observations

Limit the population model's predictive performance as well as the robustness of the statistical associations estimated between mosquito abundance and eco-climatic variables!

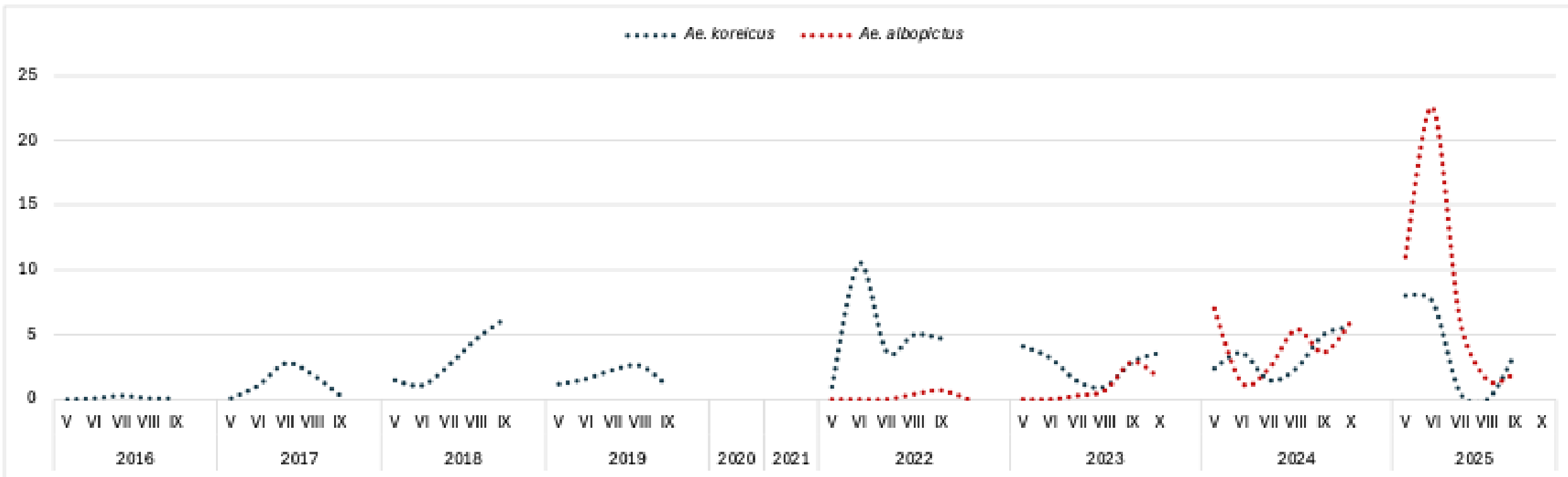
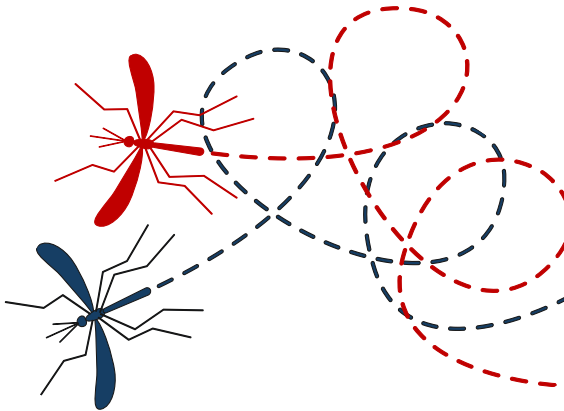




Invasion history of *Aedes koreicus* in Pécs, Hungary

Follow the temporal dynamics of *Ae. koreicus* since 2016

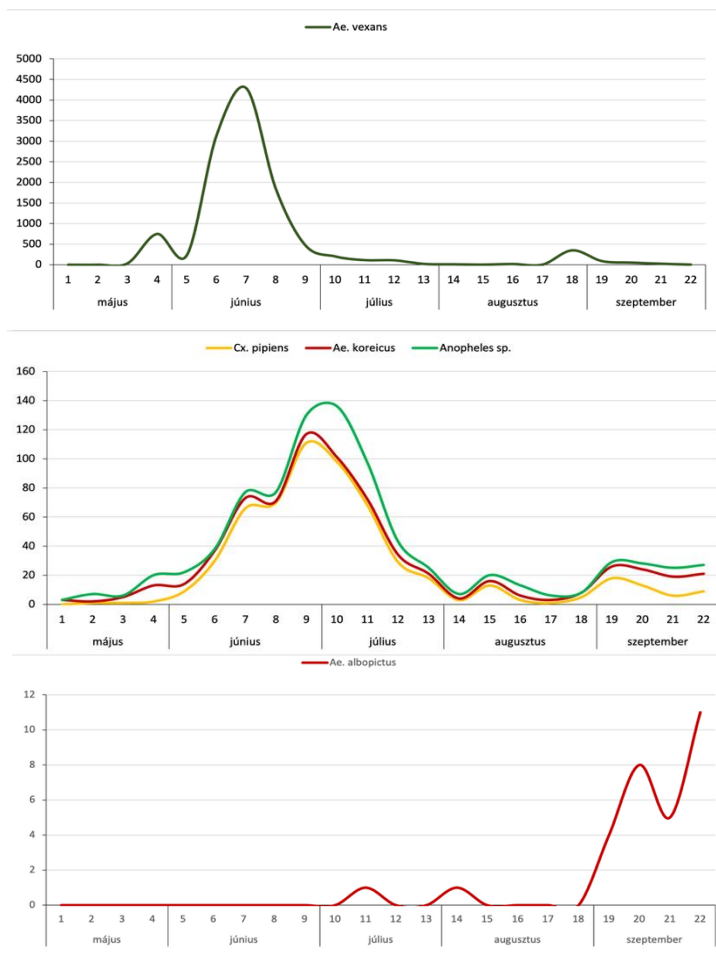
- First emergence of *Ae. albopictus* in the city of Pécs (2022?)



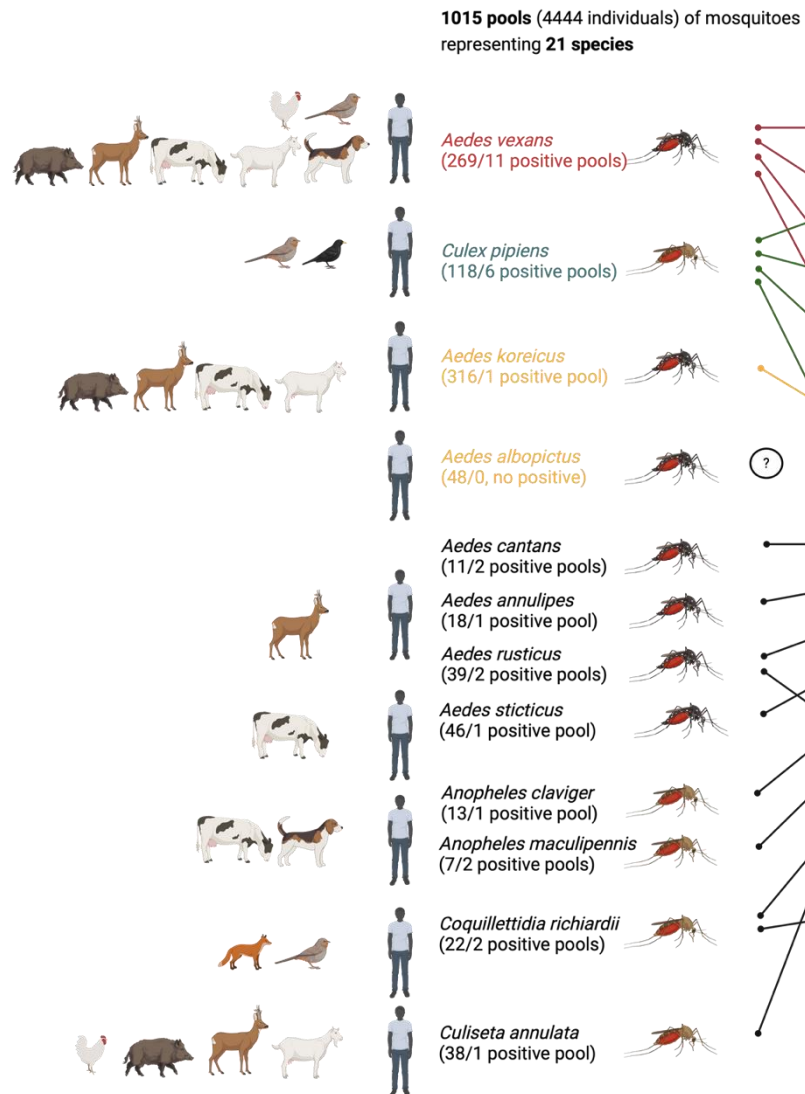


Reveal the background of human *Dirofilaria* infections

Ecology, Environment

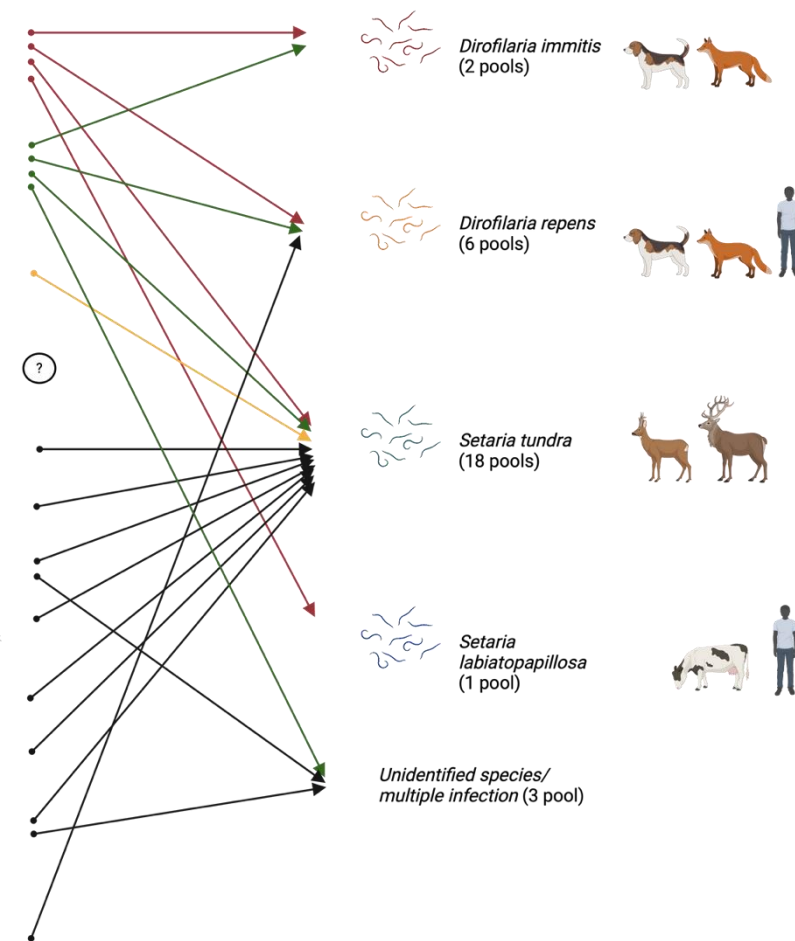


Vector behaviour



Pathogen – Host interaction

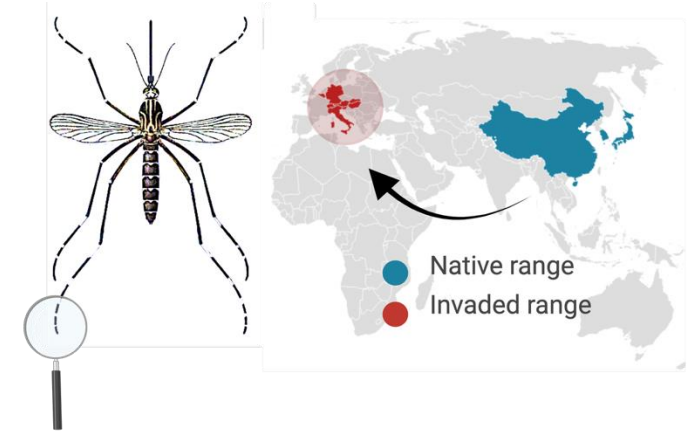
30 positive pools
(EIR = 0.007, 95% CI: 0.005 - 0.009)





What we (do not) know about *Aedes koreicus*

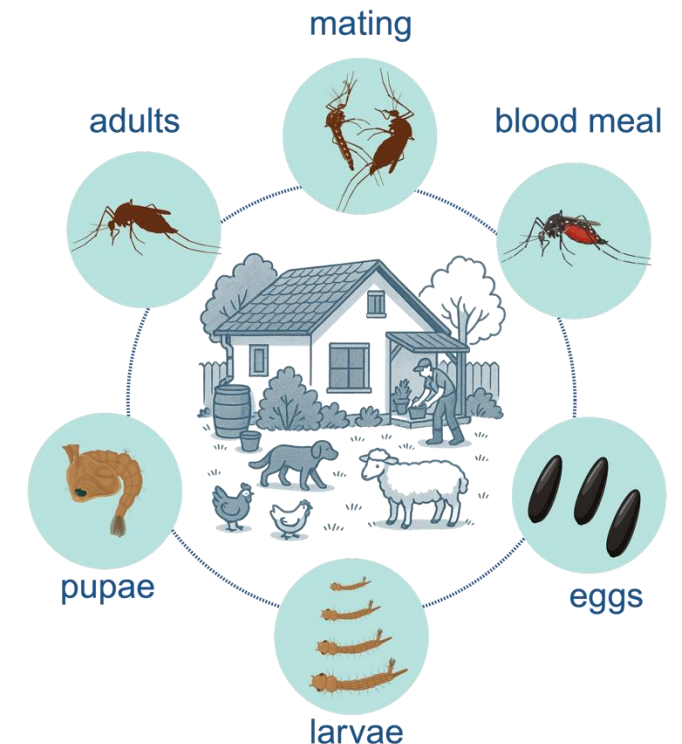
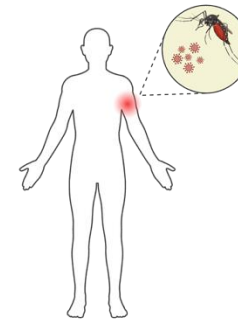
- two morphological forms are present: “Jeju-Do” and “Mainland Korea”
- prefers human-inhabited areas, but not too much
- “slowed down” dispersal and invasiveness
- feed on different kinds of animals
- known to bite humans during the daytime, but not as annoying as the tiger mosquito
- able to carry and/or transmit (biological vector) several pathogens



Increasing the potential transmission of pathogens

Potential vector of viruses and parasites

	Japanese encephalitis virus			<i>Brugia malayi</i>	
	Zika virus			<i>Dirofilaria immitis</i>	
	Chikungunya virus			<i>Dirofilaria repens</i>	
	Sindbis virus			<i>Setaria tundra</i>	





Outstanding questions

Invasive status of the species

- Seasonal dynamics
(abundance, generations per season)
- Environmental thresholds
- Possible future expansion

Vectorial capacity in the new ecosystem

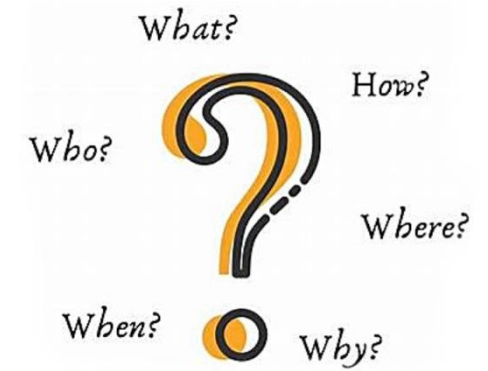
- Vector status
- Role as an enzootic or bridge vector
- Climate-sensitive pathogens
- Blood-feeding patterns and host preferences

Ecological plasticity

- Ecological niche
- Needs and preferences
- “age-dependent” adaptation
- Interactions with other invasive species

Control approaches

- Nuisance and/or Vector status
- Integration to general/AIM-specific control actions



Thank you!



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